

TECHNICAL NOTE 2

WAR GAMING AND SCENARIO PLANNING

Red Team Methodology, Shell Method, and Assumption-Based Planning

1. Strategic War Gaming: Foundations

Business war gaming adapts military simulation techniques to anticipate competitive dynamics. Unlike static analysis, war games force participants to 'think like the competitor,' revealing hidden assumptions and organisational blind spots that traditional analysis misses.

 **KEY INSIGHT:** The key insight from military applications: humans make different decisions when they must commit to actions under uncertainty than when they merely analyse situations abstractly.

1.1 Historical Origins

War gaming has roots in Prussian military tradition (Kriegsspiel, 1812) and was extensively developed during the Cold War. Business adaptation began in the 1980s when companies like Shell Oil demonstrated that simulation techniques could improve strategic decision-making.

CASE EXAMPLE: Shell Oil and the 1973 Crisis

In the early 1970s, Shell's planning team, led by Pierre Wack, developed scenarios exploring potential oil supply disruptions. When OPEC announced the oil embargo in October 1973, most oil companies were caught off guard.

Shell's advantage: Their teams had mentally rehearsed responses to this exact scenario years before it occurred. While competitors scrambled to react, Shell executives could immediately implement pre-considered strategies.

Result: Shell moved from #7 to #2 among the major oil companies within two years, demonstrating the competitive advantage of scenario-based preparation.

1.2 Why War Gaming Works

War gaming produces superior strategic insights through four key mechanisms:

Perspective Shift: Forces executives to think from competitor viewpoints, breaking home-team bias. When you must 'win' for the competitor, you discover attack vectors your defensive mindset would never consider.

Assumption Testing: Reveals unstated assumptions that underpin strategic plans. Many strategies fail not because of poor analysis but because critical assumptions prove false.

Dynamic Interaction: Captures second and third-order effects that static analysis misses. Your move triggers their response, which changes your next move—this cascading dynamic is invisible in spreadsheet models.

Safe Failure: Allows testing of risky strategies without real-world consequences. Better to discover a strategy's vulnerability in a simulation than in the marketplace.

CASE EXAMPLE: Kodak's Missed War Game

Kodak invented the digital camera in 1975. Internal war games could have revealed the existential threat digital posed to their film business. Instead, Kodak's planning treated digital as a distant, manageable threat.

A properly designed war game would have assigned a team to 'attack' Kodak's film business using digital technology. This team would have discovered:

- Film margins could not survive digital competition
- Distribution partners would abandon the film quickly
- Consumer switching would accelerate exponentially

The failure wasn't in capability (Kodak had the technology) but in assumption—they assumed digital would remain a niche product. War gaming would have explicitly challenged this assumption.

2. Red Team / Blue Team Methodology

Originating in military exercises and later adopted in cybersecurity, this technique assigns dedicated teams to attack (Red) and defend (Blue) strategic positions. The creative tension between teams produces insights neither could achieve alone.

Figure 2.1 | Red Team / Blue Team Structure

War Gaming Organizational Framework



Based on Oriesek & Schwarz (2008). *Business Wargaming: Securing Corporate Value*.

Figur

e 1.1: Red Team / Blue Team Structure

War Gaming Organisational Framework showing team roles, composition, and interaction flows.

2.1 Team Roles and Responsibilities

Red Team (Competitor Simulation)

Composition: Ideally includes people with deep knowledge of the competitor—former employees, industry analysts, or executives who have competed against them. Fresh perspectives from outside the organisation are valuable.

Information Package: Receives a comprehensive briefing on the competitor's resources, culture, historical strategy patterns, and current market situation. The better the intelligence, the more realistic the simulation.

Objective: Identify vulnerabilities in the Blue Team's strategy from an external perspective. The Red Team must genuinely try to 'win' for the competitor—softball responses destroy the exercise's value.

RED TEAM MINDSET

Critical success factor: Red Team members must psychologically commit to their competitor role. This requires:

- Suspending loyalty to the home organisation during the exercise
- Genuinely advocating for competitors' interests
- Celebrating when they find vulnerabilities in the Blue Team's strategy
- Receiving recognition for effective 'attacks' rather than being seen as disloyal

War games fail when Red Teams 'pull punches' or when organisational culture punishes those who effectively challenge the company's strategy.

Blue Team (Home Team)

Composition: Typically includes the executives who will actually implement the strategy being tested. Their participation creates ownership of insights and accelerates post-game implementation.

Information Package: Full access to internal strategic plans and resources. Blue Team operates with the same information they would have in reality.

Objective: Defend current strategy while identifying countermeasures to Red Team attacks. Must engage seriously with threats, not dismiss them as unrealistic.

Control Team (Facilitation)

Composition: Experienced facilitators, potentially external consultants for objectivity. Control Team members must have sufficient industry knowledge to evaluate the plausibility of moves.

Key Responsibilities:

- Design game structure and rules
- Introduce exogenous shocks (regulatory changes, technology disruptions, economic shifts)
- Evaluate plausibility of proposed moves
- Manage time and process
- Ensure game produces actionable insights, not just entertainment

 **KEY INSIGHT:** Debrief is critical: At least 30% of total time should be dedicated to analysing lessons learned. The game itself produces insights; the debrief captures and codifies them.

2.2 Game Structure



A typical war game follows this structure:

Pre-Game Briefing (2-4 weeks before): Teams receive information packages and prepare initial strategies. Red Team studies competitor deeply; Blue Team refines strategic plan.

Opening Moves (Round 1): Each team presents its initial strategic position. This establishes the baseline from which the game evolves.

Response Rounds (2-4 rounds): Alternating moves and counter-moves, each representing 6-12 months of simulated time. Teams have 30-60 minutes per round to develop and present their moves.

Shock Events: Control Team introduces market changes, regulatory shifts, or competitive surprises. These test the robustness of strategies under uncertainty.

Resolution: Quantitative models or facilitator judgment determine the outcomes of each round's decisions.

Debrief (Critical): Structured analysis of what happened, what surprised participants, and what actions should follow.

CASE EXAMPLE: Intel vs. AMD War Game

In the late 1990s, Intel reportedly conducted war games simulating AMD's potential response to Intel's pricing strategies. The Red Team (playing AMD) identified that Intel's dominant market share created antitrust vulnerability.

War Game Insight: Aggressive pricing to crush AMD could trigger regulatory intervention that would ultimately harm Intel more than the competitive threat.

Strategic Outcome: Intel moderated its competitive response, maintaining market dominance while managing regulatory risk. The war game revealed that 'winning too completely' was itself a vulnerability.

3. Scenario Planning: The Shell Method

Scenarios are organisational learning tools, not predictions. The Shell method provides a structured approach to developing scenarios that challenge assumptions and expand strategic thinking.

 **KEY INSIGHT:** Pierre Wack's fundamental insight: 'Scenarios are not predictions—they are tools for learning and challenging mental models. The goal is not to be 'right' about the future but to be prepared for multiple futures.'

3.1 The Five-Step Process

Step 1: Identify Driving Forces

Driving forces are factors that will shape the future regardless of what individual actors do. Systematically scan across categories:

Social: Demographic changes, cultural shifts, lifestyle trends, urbanisation patterns

Technological: Emerging technologies, digitalisation, automation, platform effects

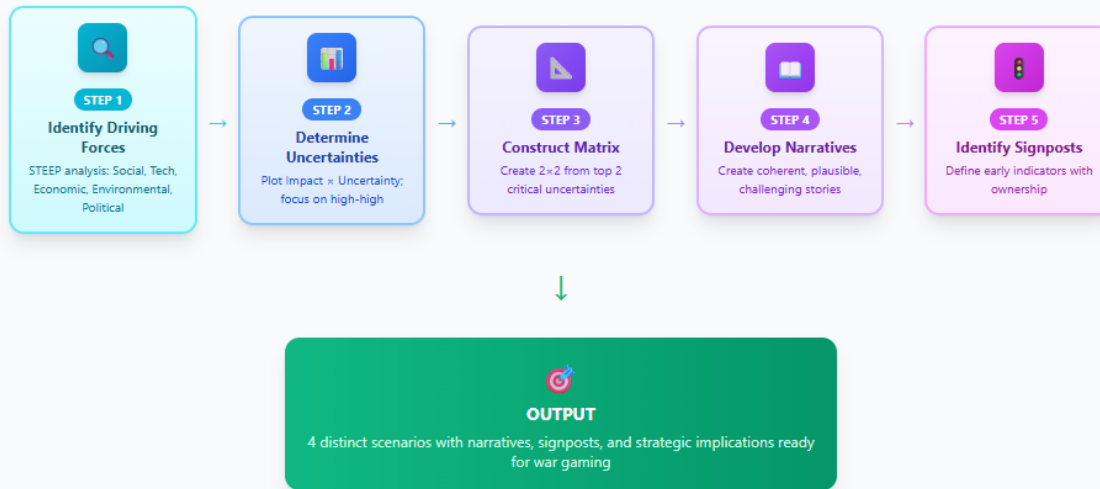
Economic: Growth rates, inflation, trade patterns, income distribution

Environmental: Climate change, resource constraints, sustainability requirements

Political: Regulatory trends, geopolitical shifts, nationalism vs. globalisation

Figure 1.3 | Shell Method Process Flow

Five-Step Scenario Planning Methodology



Pierre Wack's Key Insight

"Scenarios are not predictions—they are tools for learning and challenging mental models. The goal is not to be 'right' about the future but to be prepared for multiple futures. **The process of creating scenarios is often more valuable than the scenarios themselves.**"

Based on Wack, P. (1985). *Scenarios: Uncharted Waters Ahead*. Harvard Business Review.

Figure 2.3: Shell Method Process Flow

Five-step methodology from driving forces identification through signpost monitoring

Step 2: Determine Critical Uncertainties

Not all driving forces are equally important. Critical uncertainties are factors that are both:

Highly Uncertain: Could plausibly go in multiple directions. If you're confident about the outcome, it's a 'predetermined element,' not an uncertainty.

Highly Impactful: Would significantly affect strategic success regardless of which direction it goes.

Plot driving forces on a 2x2 matrix (Uncertainty × Impact). Focus on the high-high quadrant when constructing scenarios.

Step 3: Construct Scenario Matrix



Select the two most critical uncertainties. Create a 2x2 matrix where each quadrant represents a distinct scenario:

- Scenario A: High on both uncertainties
- Scenario B: High on first, low on second
- Scenario C: Low on first, high on second
- Scenario D: Low on both uncertainties

Each scenario must be internally consistent and plausible—not every combination makes logical sense.

Step 4: Develop Narratives

For each scenario, create a coherent story that:

- Explains how the world evolved from today to that scenario
- Is internally consistent (no logical contradictions)
- Is plausible (not science fiction)
- Is challenging (not comfortable extensions of current trends)
- Has a memorable name that captures its essence

SCENARIO NAMING EXAMPLES

Good scenario names are evocative and memorable:

- 'Green Revolution' — rapid transition to sustainable energy
- 'Fortress Europe' — regional fragmentation and protectionism
- 'Winner Takes All' — platform concentration and monopoly power
- 'Muddling Through' — incremental change, no dramatic shifts

Avoid generic names like 'Scenario 1' or 'High Growth Case'—these don't create mental hooks for future reference.

Step 5: Identify Signposts

Signposts are early indicators that signal which scenario is unfolding:

Leading Indicators: Events that precede scenario emergence

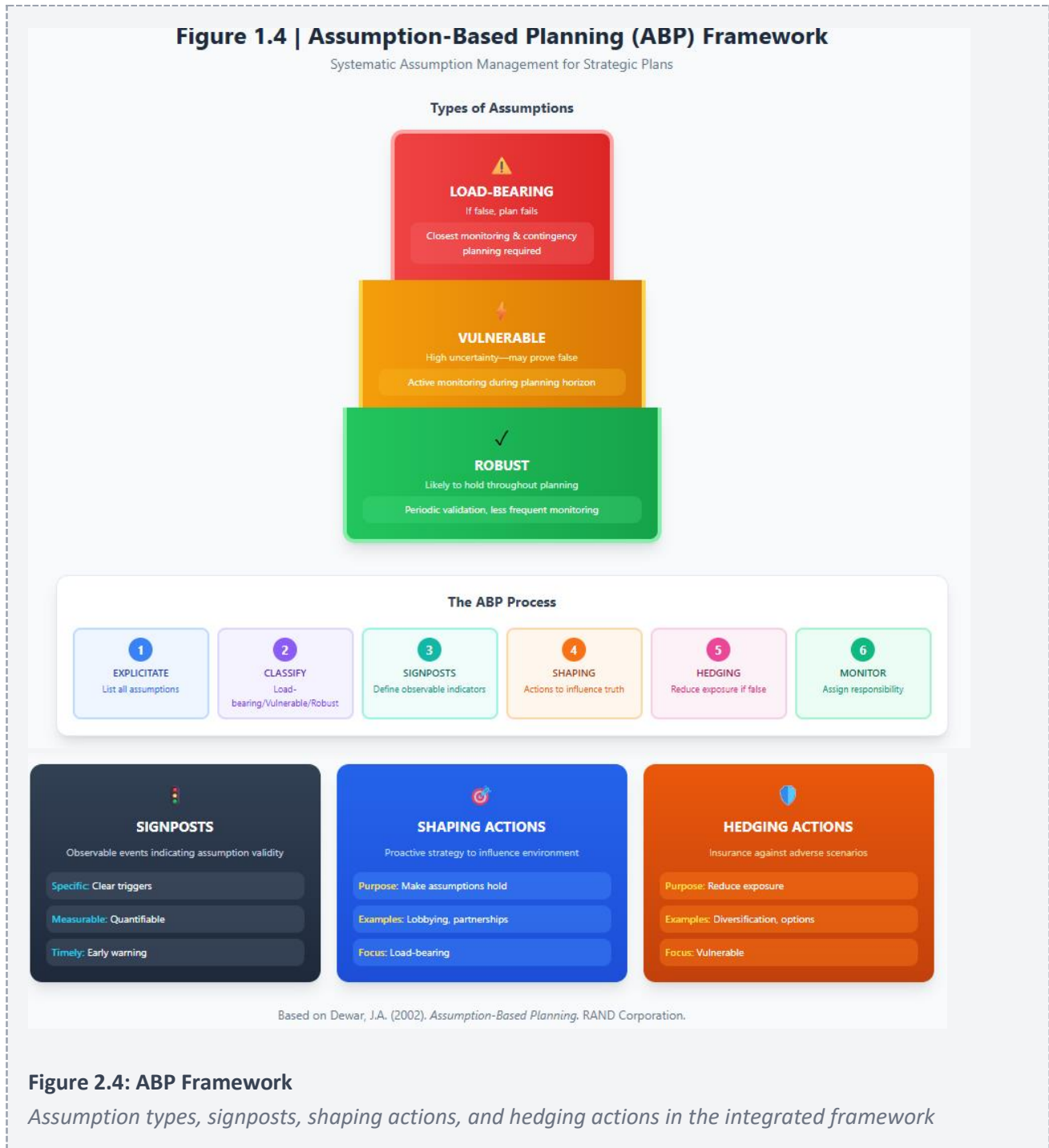
Tracking Metrics: Quantitative measures to monitor

Trigger Events: Specific occurrences that would confirm a scenario

Assign responsibility for monitoring each signpost and establish escalation protocols. Without monitoring, scenarios become academic exercises rather than living strategic tools.

4. Assumption-Based Planning (ABP)

ABP focuses on explicitly identifying and testing the assumptions underlying strategic plans. It complements scenario planning by providing a systematic framework for assumption management.



4.1 Types of Assumptions

Load-Bearing Assumptions: Assumptions that, if false, would cause plan failure. These require the closest monitoring and contingency planning. Like load-bearing walls in architecture, removing them collapses the entire structure.

Vulnerable Assumptions: Assumptions with high uncertainty may prove false within the planning horizon. Require active monitoring, but the plan can potentially adapt if they fail.

Robust Assumptions: Assumptions likely to hold throughout the planning period. Less monitoring is required, but periodic validation is needed—even 'certain' things occasionally change.

CASE EXAMPLE: Nokia's Load-Bearing Assumption

In 2007, Nokia's strategy rested on a load-bearing assumption: 'Mobile phones are primarily communication devices, and consumers choose based on call quality, battery life, and durability.'

The iPhone revealed that this assumption was false. Consumers wanted pocket computers that also made calls. Nokia's entire product strategy, supply chain, and organisational structure were built on the wrong assumption.

ABP Application: Had Nokia explicitly identified this as a load-bearing assumption and assigned signposts (smartphone app downloads, mobile web usage, consumer research on 'ideal phone'), they might have detected the shift earlier.

Key Lesson: The most dangerous assumptions are those so embedded in organisational thinking that they're invisible.

4.2 The ABP Process

1. Explicitate Assumptions: List all assumptions underlying the strategic plan. This is harder than it sounds—many assumptions are implicit and invisible to those who hold them.

2. Classify Assumptions: Categorise each as load-bearing, vulnerable, or robust based on (a) how critical it is to plan success and (b) how certain you are that it will hold.

3. Identify Signposts: Define observable events that would indicate assumption validity. Good signposts are specific, measurable, and timely.

4. Design Shaping Actions: Actions the organisation can take to influence whether assumptions hold—proactive strategy to shape the environment rather than merely react to it.

5. Design Hedging Actions: Actions that reduce exposure if assumptions prove false—insurance against adverse scenarios.

6. Establish Monitoring: Assign responsibility and frequency for tracking signposts. Without clear ownership, monitoring doesn't happen.

4.3 Signposts and Actions

SIGNPOSTS, SHAPING, AND HEDGING

SIGNPOSTS — Observable events indicating assumption validity:

- Must be specific and measurable
- Should provide early warning, not confirmation
- Need clear ownership and monitoring frequency

SHAPING ACTIONS — Proactive strategy to influence the environment:

- Lobbying for favourable regulation
- Developing partnerships that reinforce assumptions
- Market development activities

HEDGING ACTIONS — Insurance against adverse scenarios:

- Diversification into alternative business models
- Real options (investments that create future flexibility)
- Contingency plans ready for rapid deployment

5. Economic Evaluation of Scenarios

The ECOMO model provides a framework for evaluating strategic alternatives across scenarios, enabling quantitative comparison of options.

5.1 Cost Categories

Development Costs: Initial investment required for software development, system integration, training, and infrastructure.

Operating Costs: Ongoing expenses—maintenance, personnel, data acquisition, licensing.

Opportunity Costs: Value of foregone alternatives. Resources committed to one strategy cannot be used elsewhere.

5.2 Value Categories

Monetary Value: Quantifiable financial benefits—cost savings, revenue increases, risk reduction with assignable probability and impact.

Psychological Value: Non-quantifiable benefits—executive confidence, organisational learning, strategic flexibility.

5.3 Expected Value Calculation

For uncertain outcomes, use the Expected Value methodology:

Expected Value = Σ (Outcome Value $\times$ Probability of Outcome)

EXPECTED VALUE EXAMPLE

A competitive intelligence system might prevent three types of strategic errors:

Error Type A: Product launch failure

- Probability of occurrence without CI: 40%
- Probability CI prevents if it occurs: 60%
- Cost of error: €5M
- Expected Value = $\text{€}5\text{M} \times 0.40 \times 0.60 = \text{€}1.2\text{M}$

Error Type B: Missed market opportunity

- Probability without CI: 30%
- Probability CI identifies: 70%
- Value of opportunity: €8M
- Expected Value = $\text{€}8\text{M} \times 0.30 \times 0.70 = \text{€}1.68\text{M}$

Total Expected Value of CI System = $\text{€}1.2\text{M} + \text{€}1.68\text{M} + (\text{other errors}) = \text{Justification for investment}$

6. Integration: From Analysis to Action

The methodologies presented in this note—war gaming, scenario planning, and assumption-based planning—are complementary tools that work best when integrated:

Scenario Planning identifies the range of possible futures and critical uncertainties.

ABP explicates the assumptions underlying your strategy and establishes monitoring systems.

War Gaming stress-tests your strategy against intelligent competitors across scenarios.

 **KEY INSIGHT:** The most effective organisations cycle through these methodologies iteratively: scenarios inform assumptions, assumptions shape war game design, war game insights refine scenarios.

6.1 When to Use Each Tool

- Use Scenario Planning when facing high uncertainty about the external environment (markets, regulation, technology).
- Use ABP when implementing any significant strategic initiative to identify hidden vulnerabilities.
- Use War Gaming when competitive dynamics are central to strategic success—when 'what competitors do' matters as much as 'what the market does.'

7. References

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